

ABHISHEK MANOJ SUTARIA

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EDUCATION

Indiana University Bloomington

Aug 2024 - May 2026

Master of Science in Data Science

GPA 3.72/4.0

Relevant Coursework: Big Data in Economics, Microeconomic Theory-I, Applied Machine Learning

University of Mumbai

Jun 2018 - May 2022

B.E. in Electronics and Telecommunication Engineering

GPA 3.9/4.0

Honors Specialization in Artificial Intelligence and Machine Learning

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Applied Statistics

TECHNICAL SKILLS

Languages	: Python, R, Spark (PySpark), SQL, Java, Shell Scripting, Scala
Libraries	: Numpy, Pandas, Matplotlib, Seaborn, Scikit-learn, Polars, NLTK, Keras, TensorFlow
Data Science	: Machine Learning, Deep Learning, Data Analytics, Business Intelligence
Technologies	: Git, MySQL, PostgreSQL, Snowflake, Power BI, Tableau, Hadoop, PyTorch, Excel
Cloud Computing	: AWS (EC2, S3), GCP

WORK EXPERIENCE

Edelweiss Global Markets, India | *Data Engineer*

Jul 2022 – Aug 2024

- Developed Python-based data quality algorithms for financial data streams, improving data accuracy by 70% and reducing ingestion errors by 40% in AWS-hosted pipelines.
- Engineered 40+ automated ETL pipelines using Airflow and Spark to process over 1TB per day of equities and futures data, supporting research for quant strategies.
- Built a validation framework in Python using Polars and Great Expectations on AWS S3 to ensure tick-by-tick data quality, boosting integrity from 85% to 98% across 500K+ records daily.
- Integrated multi-source market data feeds and trading signals via Python-based APIs, reducing cross-system latency by 40% for quant research workloads.
- Automated tick-level 10 to 12 TB daily data cleanup post-validation using shell scripts on AWS S3, saving 8+ hours weekly and improving storage utilization by 30%.

Digbi Health, California | *Data Science Intern*

Jun 2025 – Nov 2025

- Developed production-ready APIs using FastAPI, exposing cleaned and tagged datasets to downstream analytical services and internal user dashboards.
- Engineered robust data stratification logic using Python and SQL to identify chronic condition cohorts and CPT code patterns, analogous to order book-level tagging in market microstructure analysis.
- Integrated LLMs and autonomous agents to automate pipeline logic, dynamically parsing schema variants and optimizing unstructured data ingestion—cutting preprocessing time by 40%.
- Collaborated directly with CEO and engineering team to align technical outcomes with business priorities.

Indiana University Indianapolis, Indiana | *Research Intern*

May 2025 – Jul 2025

- Designed a scalable generative modeling pipeline using CycleGAN, VAE-RNN, and Diffusion models to predict high-resolution temporal changes in complex structured data, leveraging techniques relevant to modeling sequential dependencies and state transitions.
- Deployed end-to-end AI workflows in a containerized, cloud-ready architecture using Docker and PyTorch Lightning, ensuring reproducibility, version control, and performance integrity across evolving data environments.

Indiana University Bloomington, Indiana | *AI Research Analyst*

Sep 2025 - Present

- Engineered an AI-powered digital twin prototype of a university professor by orchestrating multi-agent frameworks such as CrewAI and LangChain with retrieval-augmented generation pipelines over multi-modal academic datasets, enabling contextual Q&A and personalized tutoring.
- Developed a real-time multimodal interaction system by integrating voice cloning technologies like Whisper, VALL-E, and ElevenLabs with lip-synced avatar rendering using WebRTC, React, Tailwind, and framer-motion, reducing conversational latency and delivering a hyper-realistic, human-like teaching experience.
- Optimized model alignment and retrieval workflows through knowledge distillation, reinforcement learning feedback loops, and vector database optimization using Pinecone, FAISS, and Weaviate, improving response accuracy and engagement quality in multi-turn conversations.

ACADEMIC PROJECTS AND RESEARCH PAPERS

Diabetic Retinopathy Detection using Convolution Neural Networks and DR-GANs

Jun 2021

- Designed a deep learning system using CNNs and GANs to automate diabetic retinopathy classification, achieving 92.3% accuracy across DR severity stages.
- Collaborated on model development using Git and GitHub for version control, branching, and code reviews, enabling efficient teamwork and scaling processing to 1,000 images/hour, improving diagnostic speed 10x.
- Reduced diagnosis time by 60%, facilitating earlier clinical intervention and reducing risk of vision loss.
- Secured 1st place in IETE-SF DJ Spark technical paper competition and published research at the state level.

Elements of Nature Optimized into Smart Energy Grids using Machine Learning

Jan 2022

- Developed a solar radiation forecasting model using hyperparameter-tuned Random Forests, achieving 72% accuracy and outperforming traditional methods by 25%.
- Implemented Lasso, SVM, and Linear Regression algorithms, and validated performance using ROC curves and confusion matrices to support smart grid decisioning.
- Conducted data cleaning and feature engineering on weather station datasets using Pandas and correlation analysis, identifying temperature as the most influential predictor of solar output.
- Published in the Design Engineering Journal; the paper has been cited over 82 times, reflecting its technical relevance and academic impact in the field of smart energy systems.

FinEdu.ai – Finance Education Q&A LLM Assistant

Nov 2024

- Designed and implemented an LLM (Finedu.ai) by integrating the Llama-2-7b-chat-hf model with RAG, Sentence-Transformers, and LlamaIndex, enabling accurate retrieval of financial definitions and delivering beginner-friendly explanations that improved learning accessibility.
- Built and deployed an interactive Gradio-based interface that ingests Kaggle financial terminology datasets, vectorizes them with HuggingFace embeddings and Torch, and performs semantic search through LangChain pipelines, resulting in clear, jargon-free responses tailored for finance students and novice investors.

MeetSmart – AI-Powered Meeting Companion

Mar 2025

- Architected and deployed “MeetSmart,” an AI-powered meeting intelligence platform leveraging FastAPI, Whisper ASR, and PyAudio for real-time multilingual speech-to-text, abstractive/extractive NLP summarization, and VADER sentiment analysis—cutting post-meeting documentation time by 80%.
- Engineered modular NLP pipelines with regex-based action item extraction, Google Translate API for multi-language support, Matplotlib and WordCloud visualizations, and SMTP-based automated email distribution—improving accuracy of action tracking more than 90% and streamlining collaboration across distributed teams.